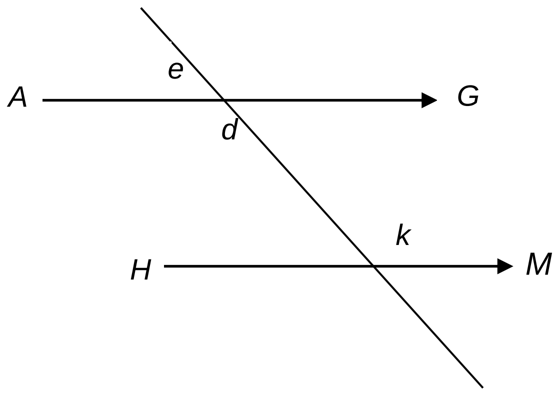
Angle properties related to parallel or perpendicular lines

Teacher resource

Question 1

In the diagram below, lines *AG* and *HM* are parallel.



If angle *d* is four times the size of angle *e*, what is the size of angle *k*? __________

Answer: 144°
Skill: Students use properties of angles on parallel lines.

Background information

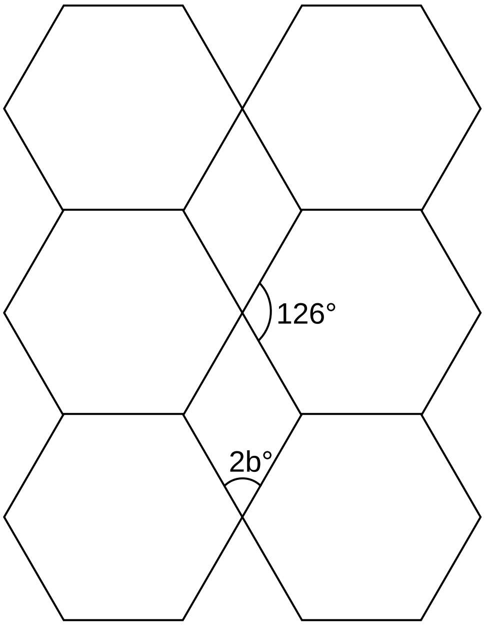
Students should understand that when a transversal crosses a pair of parallel lines the angles created are related to each other. By knowing one of these angles it is possible to work out all the other angles. Some secondary students may need practical opportunities to explore the relationships between the angles before they are well placed to calculate missing angles on a diagram. They should also understand that a diagram, which is not drawn to scale, can carry information about angles without the angles on the diagram actually measuring the size indicated. A diagram is not the same as an accurate geometric drawing.

Additional questions

1. Eight angles are formed when a transversal crosses a pair of parallel lines.
   What is the size of each the angles in the diagram, expressed in terms of *e*?
2. What would be the size of each the eight angles if angle *d* was five times the size of angle *e*?
3. What would be the size of each of the angles, expressed in terms of *d*, if angle *d* was five times the size of angle *e*?

Question 2

In this diagram, the opposite sides of each quadrilateral are parallel.



What is the value of *b*?

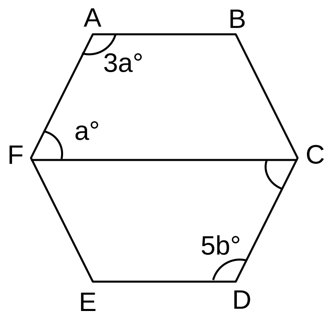
b° = ___________

Answer: 27°
Skill: Students use properties of angles of 2D shapes.

Additional questions

1. The hexagons in the diagram are all identical. Are they regular hexagons?
2. Which other angles in the diagram must also have the size of 2b°?

Question 3



In this diagram, the opposite sides of the hexagon are parallel and the diagonal FC is parallel to side ED.

What is the value of b?

b° = __________

Answer: 27°
Skill: Students use the relationships between angles created when a transversal crosses a pair of parallel lines.

Additional questions

1. Is this a regular hexagon?
2. Draw a diagram of this hexagon but do not show the actual sizes of the angles. Use conventional geometric symbols to show the parallel sides and equal angles.

Question 4

ABCDEF is an irregular convex hexagon with three pairs of parallel sides.

What is the size of angle *D* if the size of angle *A* is 62º and the size of angle *C* is 155º?

____

Answer: 62º
Skill: Students apply knowledge of polygons and angles between a transversal and parallel lines.

Additional questions

1. Calculate the sizes of the other angles.
2. Draw a diagram of this hexagon but do not show the size of the angles. Use conventional geometric symbols to show parallel sides, equal sides and equal angles.

Student activity: Angle properties related to parallel or perpendicular lines

Question 1

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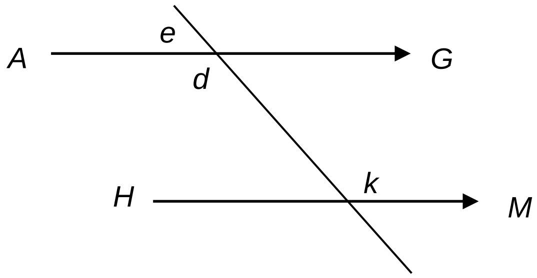

If angle *d* is four times the size of angle *e, w*hat is the size of angle *k*?

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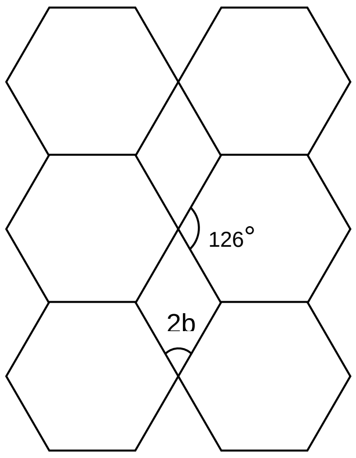
Question 2

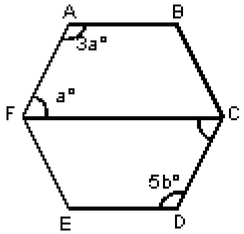
In this diagram, the opposite sides of each quadrilateral are parallel.



What is the value of *b*?

b° = ___________

Question 3



In this diagram, the opposite sides of the hexagon are parallel and the diagonal FC is parallel to side ED.

What is the value of b?

b° = ___________

Question 4

ABCDEF is an irregular convex hexagon with three pairs of parallel sides.

What is the size of angle *D* if the size of angle *A* is 62º and the size of angle *C* is 155º?

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